

Group Members:

Shang Shi - 101173744

Edward Anokye - 101220561

Ka Hei Chan - 101260193

Question 4(a)

The gradient of the objective function is given by:

$$\frac{\partial f}{\partial a} = (ax_1 + b - y_1)x_1 + (ax_2 + b - y_2)x_2$$

$$\frac{\partial f}{\partial b} = (ax_1 + b - y_1) + (ax_2 + b - y_2)$$

Using gradient descent with initial guess $(a, b) = (0, 0)$ and learning rate $\alpha = 0.01$, the algorithm converged to:

$$a = 0.24999953, \quad b = -0.24999479$$

with minimum value:

$$f(a, b) = 2.62 \times 10^{-11}.$$

This matches the analytical least-squares solution.

Question 4(b)

The gradient was derived using the chain rule and the derivative of the SiLU activation function:

$$\text{SiLU}'(x) = \sigma(x) + x\sigma(x)(1 - \sigma(x)),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function.

Using gradient descent with initial guess $(a, b) = (0.1, 0.1)$ and learning rate $\alpha = 0.01$, the algorithm converged to:

$$a = 0.39484, \quad b = -0.44204$$

with minimum value:

$$f(a, b) = 0.11129.$$

Unlike the linear case, this objective function is non-convex and does not admit a closed-form solution; therefore, gradient descent was required.